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THE SINGLE ENTITY PART A

ISE305 – DATABASE DESIGN & PRACTICE

SOME SLIDE MATERIAL FROM WATSON

Reading

- Wikipedia
  - Data Model
  - CamelCase
- Textbook – Chapter 3, pages 57-71 (lightly read "Defining a table" and "Inserting rows into a table")

Reference

- MS Access data types

  msdn.microsoft.com/en-us/library/office/ff193793.aspx

Session Objectives

- Understand the background of the relational model
- Learn to
  - Model a single entity
  - Define a single database
  - Learn some SQL operations
  - Write queries for a single-table database

Relational Model

- Basis for most DBMS and databases in use today (and likely well into the future)
- Strong mathematical foundation
  - Attributed to Codd (1970)
  - Relational algebra
  - Related to mathematical concept of a set
- A relational database is a collection of relations (mathematical term for table), where
  - All the rows of a table store the same type of item
  - Each row is uniquely identified

Relational Model History

- 1970 – proposed by Codd (considered the most fundamental result in Computer Science)
- 1976 – System R (IBM)
- Structured Query Language (SQL)
- 1980s – Commercial relational DBMSs
- 1990s – Dominance of relational approach
- 2000s – Development of object relational mapping approaches

What is a Data Model

- Integrated collection of concepts for describing data, relationships between data, and constraints on the data used by an organization.
- Has three components:
  - a structural part;
  - a manipulative part;
  - a set of integrity rules.

Terminology

- **Relation**: table with columns and rows
- **Attribute (or property)**: named column of a relation
- **Domain**: set of allowable values for one or more attributes
- **Tuple**: a row of a relation
- **Relational Database** - collection of normalized relations with distinct relation names

Modeling Reality

- A database must mirror the real world if it is to answer questions about the real world
- Data modeling is a design technique for capturing reality
  - Object modeling
  - Entity-Relationship modeling

ER and OO modeling are using a more consistent notation

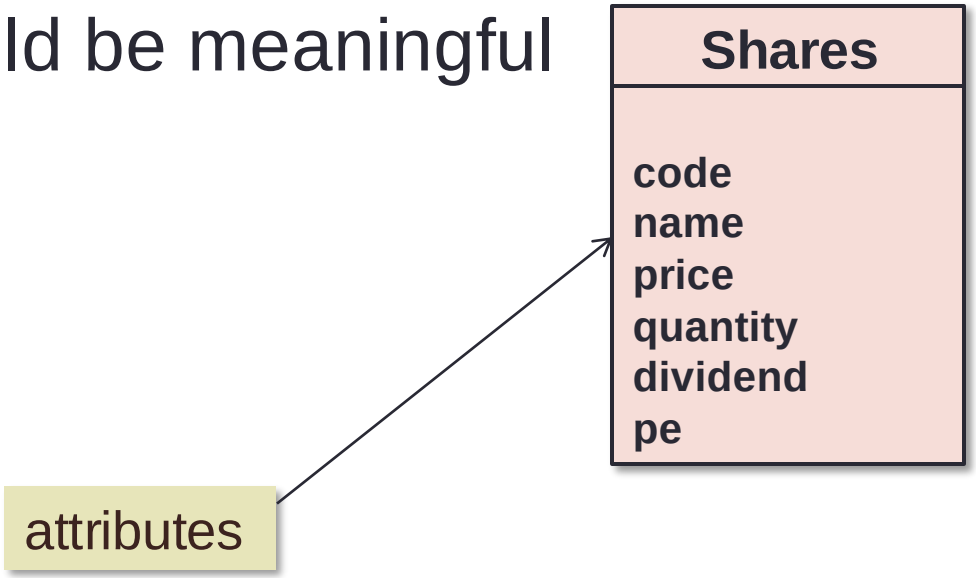
Data modeling is important for IS students

An Entity

- Usually corresponds to something concrete in the domain of the application
- Represented by a rectangle
- An instance is a particular occurrence of an entity (corresponds to a row in a DB table)

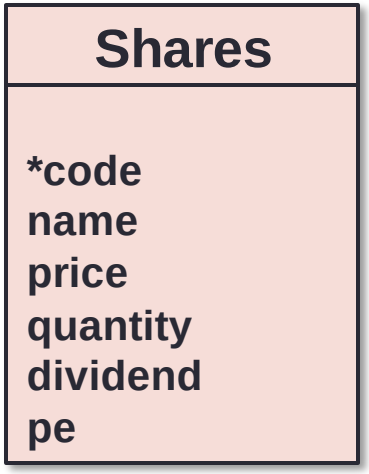
Attributes

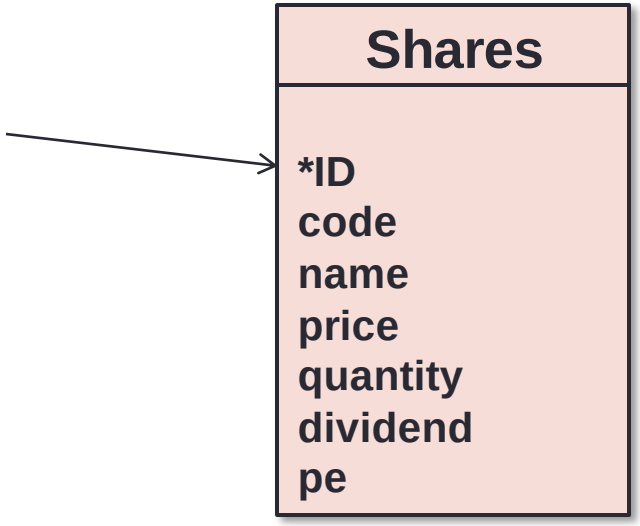
- Also referred to as properties
- An attribute is a discrete data element that describes an entity
- Attribute names should be meaningful



Identifiers (Primary Keys)

- Every instance of an entity (think row of a table) must be uniquely identified
- An identifier (primary key) can be one or more attributes
- Better to use an identifier that does not relate to a domain attribute (guaranteed uniqueness)
- A leading asterisk denotes an identifier (sometimes other notations is used, e.g., PK)



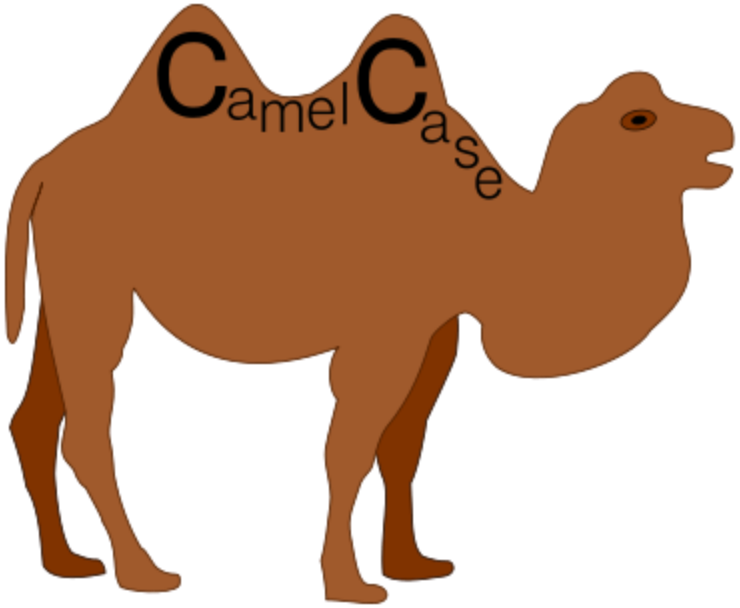


DB Naming Conventions

- No universal standard
- Good to be consistent within a project
- Camel case used frequently

CamelCase

- A compound word begins each element with a capital letter
  - Upper camel case (UCC)
  - Lower camel case (LCC) - first letter not capitalized
- Examples
  - UCC – “CamelCase”
  - LCC – “camelCase”



ISE305 DB Naming Conventions

- Camel case for tables (upper cc) and columns (lower cc)
- Table names – plural (unlike OO convention)
- Column names – singular
- Primary key field – ID
- Avoid acronyms and abbreviations except where well known (e.g., PI for Principal Investigator)

Textbook follows a slightly different naming convention

Are We on Track?

- Modify your SBU_Awards DB so that it conforms with the ISE305 DB naming conventions

Were We on Track?

Columns in Awards table, so “awd” is not needed in most attribute names

- ID => ID
- Awd Number => awardNumber or awdNum or …
- Awd Pi Full Name => PIName or piName
- Awd Type => type
- Awd Organization => organization
- Etc.

Defining a Table With Access

Shares

Field Name	Data Type
ID	Number
code	Text
shareName	Text
price	Number
quantity	Number
dividend	Number
pe	Number

Notice that attributes have a type

General | Lookup

Field Size	Double
Format	Standard
Decimal Places	Auto
Input Mask	
Caption	
Default Value	
Validation Rule	
Validation Text	
Required	No
Indexed	No
Smart Tags	
Text Align	General

Details of the type are contained in the General pane

Are We on Track?

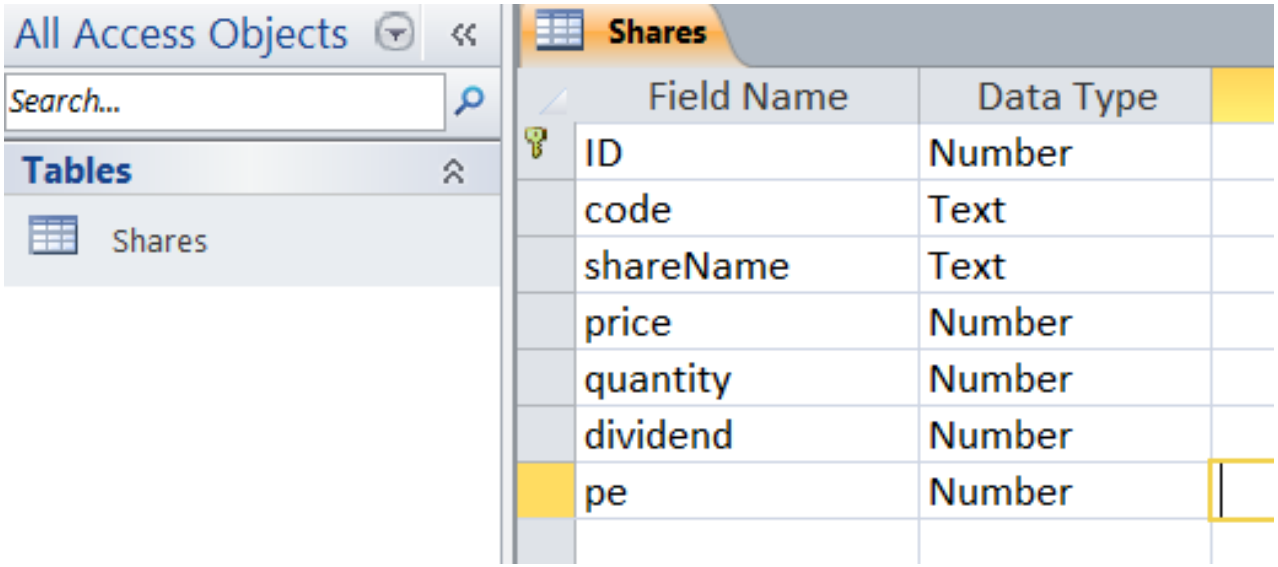
- Create a new DB from the Shares.xlsx file posted on Blackboard.
  - Use the spreadsheet column names as your attribute names
  - Use the default types (Text and Double)
  - Choose the ID attribute as your primary key
  - Do not use the Analyze Wizard
- Organize
  - Resize the columns
  - Look at table in MS Access Design View

Shares.xlsx contains the data used in the Chapter 3 textbook examples

Double in Access has a type of Number and field size property of Double

Were We on Track?

- MS Access Design View



Data Type Recap

- Covered in ISE208
- Definition - a classification identifying one of various types of data that determines:
  - the possible values for that type
  - the operations that can be done on values of that type
  - the meaning of the data
  - and the way values of that type can be stored
- Issues
  - Implementation in hardware (porting issues)
  - Variations in programing languages (e.g., SQL and Java)

Definition - Wikipedia

SQL Data Type Categories

- Character strings – includes fixed and variable length plus an option for international character sets
- Bit strings – fixed and variable length
- Integer (regular and short)
- Floating point (regular and double precision)
- Numeric or decimal
- Date – includes date, time, date/time, plus timestamp and time zone

Allowable Data Types

- SQL standard

Numeric	integer	A 31-bit signed binary value
	smallint	A 15-bit signed binary value
	float(*p*)	A scientific format number of *p* binary digits precision
	decimal(*p*,*q*)	A packed decimal number of *p* digits total length; *q* decimal places to the right of the decimal point may be specified
String	char(*n*)	A fixed length character string of *n* characters
	varchar(*n*)	A variable length character string up to *n* characters
	text	A variable-length character string of up to 65,535 characters
Date/time	date	Date in the form *yyyymmdd*
	time	Time in the form *hhmmss*
	timestamp	A combination of date and time to the nearest microsecond
	time with time zone	Same as time, with the addition of an offset from UTC of the specified time
	timestamp with time zone	Same as timestamp, with the addition of an offset from UTC of the specified time

MS Access Data Types

BINARY	1 byte/char	Any type of data may be stored in a field of this type.
BIT	1 byte	Yes and No fields that contain only one of two values.
TINYINT	1 byte	An integer value between 0 and 255.
MONEY	8 bytes	A scaled integer
DATETIME	8 bytes	A date or time value between the years 100 and 9999.
REAL	4 bytes	A single-precision floating-point value.
FLOAT	8 bytes	A double-precision floating-point value.
SMALLINT	2 bytes	A short integer between – 32,768 and 32,767.
INTEGER	4 bytes	A long integer between – 2,147,483,648 and 2,147,483,647.
DECIMAL	17 bytes	An exact numeric data type
TEXT	2 bytes/char	
IMAGE	As required	Zero to a maximum of 2.14 gigabytes.
CHARACTER	2 bytes/char	Zero to 255 characters.

Access GUI uses some synonyms

Are We on Track?

- Using the Shares table you just created
  - Select more appropriate types for those attributes whose type can be improved

Were We on Track?

- Possible improvements to Shares table
  - Limit field size of code to 4
  - Change type of price attribute to Currency (2 decimal places)
  - Change format of quantity to Number
  - Change type of dividend to Currency (2 decimal places)
  - Change decimal places of pe to 0

ID	code	shareName	price	quantity	dividend	pe	Cl
1	FC	Freedonia Copper	$27.50	10,529.00	$1.84	16	
2	PT	Patagonian Tea	$55.25	12,635.00	$2.50	10	
3	AR	Abyssinian Ruby	$31.82	22,010.00	$1.32	13	
4	SLG	Sri Lankan Gold	$50.37	32,868.00	$2.68	16	

Inserting Rows With Access

- In Access, you can use the Access GUI to insert rows (similar to Excel)

Extracting Information

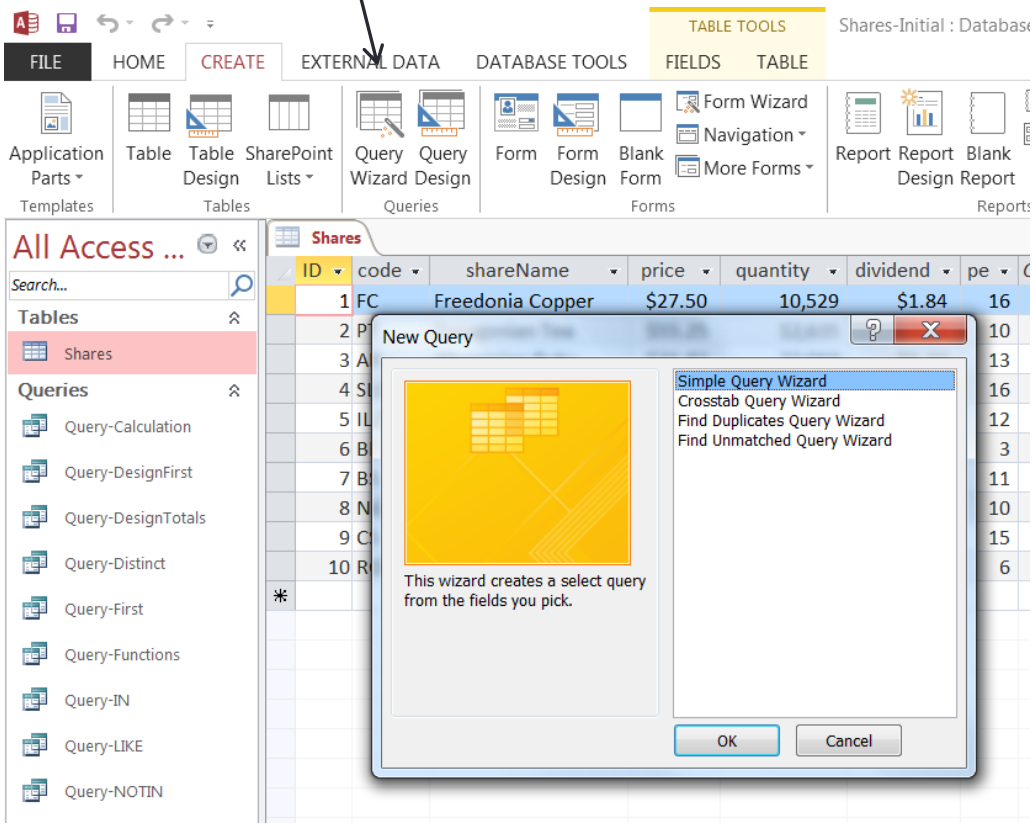
For a single table DB, the extracted data will be a collection of row subsets

- Use the SQL SELECT statement to:
  - Specify the subset of data you extract
  - Specify the ordering and grouping of data
- SELECT statement
  - Specifies the result of the extraction, not how it is to be done
  - Has optional clauses for WHERE, GROUP BY, HAVING, ORDER BY, and AS
- The DBMS translates the query into a query plan

SQL Query

- In Access, you can create a query using
  - SQL View (from an existing query)
  - Design View
- We start with the Query Wizard
  - Select fields
  - Open query at end of Wizard
  - Modify SQL

Query wizard



Querying a Table Using SQL

List all data in the share table.

```
SELECT * FROM Shares;
```

The * means *all*

Access shows the result of executing the query

ID	code	shareName	price	quantity	dividend	pe
1	FC	Freedonia Copper	$27.50	10,529.00	$1.84	16
2	PT	Patagonian Tea	$55.25	12,635.00	$2.50	10
3	AR	Abyssinian Ruby	$31.82	22,010.00	$1.32	13
4	SLG	Sri Lankan Gold	$50.37	32,868.00	$2.68	16
5	ILZ	Indian Lead & Zinc	$37.75	6,390.00	$3.00	12
6	BE	Burmese Elephant	$0.07	154,713.00	$0.01	3
7	BS	Bolivian Sheep	$12.75	231,678.00	$1.78	11
8	NG	Nigerian Geese	$35.00	12,323.00	$1.68	10
9	CS	Canadian Sugar	$52.78	4,716.00	$2.50	15
10	ROF	Royal Ostrich Farms	$33.75	1,234,923.00	$3.00	6

Note that the SQL operation produces a table

SQL Project Operation

SELECT shareName, pe FROM Shares;

Shares | SelectAll | **SelectFirmPe**

shareName	pe
Freedonia Copper	16
Patagonian Tea	10
Abyssinian Ruby	13
Sri Lankan Gold	16
Indian Lead & Zinc	12
Burmese Elephant	3
Bolivian Sheep	11
Nigerian Geese	10
Canadian Sugar	15
Royal Ostrich Farms	6

Project operation forms a new table from the identified columns

Notice that SQL keywords are displayed in capital letters

Restrict Operation

- The restrict operation creates a new table where each row in the new table satisfies a condition specified in the SQL statement

```
SELECT * FROM Shares WHERE pe < 12;
```

Shares | SelectAll

ID	code	shareName	price	quantity	dividend	pe
1	FC	Freedonia Copper	$27.50	10529	$1.84	16
2	PT	Patagonian Tea	$55.25	12635	$2.50	10
3	AR	Abyssinian Ruby	$31.82	22010	$1.32	13
4	SLG	Sri Lankan Gold	$50.37	32868	$2.68	16
5	ILZ	Indian Lead & Zinc	$37.75	6390	$3.00	12
6	BE	Burmese Elephant	$0.07	154713	$0.01	3
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11
8	NG	Nigerian Geese	$35.00	12323	$1.68	10
9	CS	Canadian Sugar	$52.78	4716	$2.50	15
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6

SQL Restrict Operation

SQL comparison operators are standard and are located in textbook, page 70

```
SELECT * FROM Shares WHERE pe < 12;
```

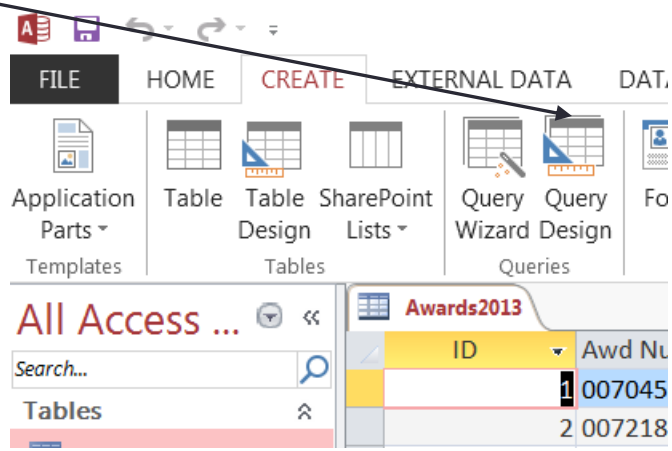
Result of query execution

Shares | SelectAll | SelectFirmPe | **SelectWhere**

ID	code	shareName	price	quantity	dividend	pe
2	PT	Patagonian Tea	$55.25	12635	$2.50	10
6	BE	Burmese Elephant	$0.07	154713	$0.01	3
7	BS	Bolivian Sheep	$12.75	231678	$1.78	11
8	NG	Nigerian Geese	$35.00	12323	$1.68	10
10	ROF	Royal Ostrich Farms	$33.75	1234923	$3.00	6
(New)						

Are We on Track?

- Create and run a SQL query on the SBU_Awards DB
  - List the budget and award name for all awards with a budget greater than $1,000,000.
- Hints
  - Open in Query Design and switch to SQL View
  - Save the query with an appropriate name (e.g., Sponsors)
  - Run the query



Are We on Track?

- Result

Awards | LargeAwards

budget	awdName
1,073,203.00	SURFACE CHEMISRTY OF PYRITE AND ISOSRUCTURAL ANALOGS: A COMBINNED SUF
1,745,291.00	DEVELOPMENT OF ACCURATE FORCING DATA IN SUPPORT OF SINGLE-COLUMN MC
2,565,289.73	RESEARCH IN NUCLEAR ASTROPHYSICS: SUPERNOVAE AND NEURON STARS
7,447,012.64	NUCLEAR REACTION STUDIES
14,837,783.09	RESEARCH IN THEORETICAL NUCLEAR PHYSICS
19,021,937.47	WATER CERENKOV DETECTOR EXPERIMENTS AND EXPERIMENTS AT HIGH ENERGY H
1,264,107.62	STUDY COMPARING SERZONE TO CBT-CD & COMBINED SERZONE & CBT-CD TREATM
8,226,393.32	Studies of Relativistic Heavy Ion Collisions at RHIC
4,938,648.81	NEURODEVELOPMENTAL & NEUROLOGICAL STUDY OF INFANTS & CHILDREN HIV-1
2,795,201.00	Variability Between Individuals with Respect to Cognitive and Physiological Resilience
1,542,546.00	Rapid Detection and Identification of Zoonotic Pathogens
1,137,828.66	NICHD International Pediatric and Perinatal HIV Studies Network
1,040,370.69	Gynecologic Oncology Group Support - CTSU - 70 Study 213 & 218
1,240,426.00	NONINVASIVE DIAGNOSIS OF EARLY BLADDER CANCERS

Were We on Track?

- SQL

  ?

Did You Meet Session Objectives?

- Learn to
  - Model a single entity
  - Define a single database
  - Learn some SQL operations
  - Write queries for a single-table database